

**Team members:**

Consider the following artificial dataset for gym memberships:

| id  | first_name | visit_frequency | signup_date |
|-----|------------|-----------------|-------------|
| 101 | Lee        | high            | 01/03/2026  |
| 102 | Aaliyah    | med             | 05/30/2026  |
| 103 | EMILY      | high            | 02/15/2026  |
| 104 | Sam        | low             | 08/10/2026  |
| 105 | omar       | med             | 09/22/2026  |

**Question 1**

For each column, write the variable type and the most appropriate R storage type.

- `first_name`:
- `visit_frequency`:
- `signup_date`:

**Question 2**

You should have identified that one of the columns would be best stored as a **factor**. Why would it be better stored as a **factor** rather than a **character**?

**Question 3**

You should have identified that one of the columns would be best stored as a **date**. Why would it be better stored as a **date** rather than a **character**?

**Question 4**

You should have identified that one of the columns would be best stored as a **character**. Why would it be better stored as a **character** rather than a **factor**?

**Question 5**

Which of the columns from the original dataset could be described as a “string”?

**Question 6**

Complete the following R code to change each of the columns to the ideal R storage type.

```
customers <-  
  customers |>  
  mutate(  
    first_name =  
  
    visit_frequency =  
  
    signup_date =  
  )
```

**Question 7**

Oh no! It looks like names were not recorded with consistent capitalization. We want them to be recorded title case, meaning the first letter should be capital, followed by lower case letters. Write R code to store the names in title case. (Hint: use the `str_to_title()` function). Make sure to save the change back into the data frame.

**Question 8**

By default, variables stored as factors will have the levels stored in alphabetical order. Complete the R code below to convert the `visit_frequency` variable to a factor with levels ordered by “low”, “med”, “high”.

```
customers <-  
  customers |>  
  mutate(  
    _____ = fct_relevel(  
      _____,  
      _____,  
      _____,  
      _____  
    )  
  )
```

**Question 9**

Actually, I decided I want the reverse order for `visit_frequency`, from high to low. Complete the R code below.

```
customers <-  
  customers |>  
  mutate(_____ = _____(_____))
```

**Question 10**

Describe what the following code seems to be doing.

```
customers |>  
  mutate(membership_length = ymd("2026-09-28") - signup_date) |>  
  arrange(membership_length)
```

|   | id  | first_name | visit_frequency | signup_date | membership_length |
|---|-----|------------|-----------------|-------------|-------------------|
| 1 | 105 | omar       | med             | 2026-09-22  | 6 days            |
| 2 | 104 | Sam        | low             | 2026-08-10  | 49 days           |
| 3 | 102 | Aaliyah    | med             | 2026-05-30  | 121 days          |
| 4 | 103 | EMILY      | high            | 2026-02-15  | 225 days          |
| 5 | 101 | Lee        | high            | 2026-01-03  | 268 days          |

**Question 11**

Make a new column called `membership_month` which stores the month that the customer signed up. (Hint: use the `month()` function.)

| id  | first_name | visit_frequency | signup_date | id  | monthly_price |
|-----|------------|-----------------|-------------|-----|---------------|
| 101 | Lee        | high            | 2026-01-03  | 100 | 15            |
| 102 | Aaliyah    | med             | 2026-05-30  | 101 | 15            |
| 103 | EMILY      | high            | 2026-02-15  | 102 | 25            |
| 104 | Sam        | low             | 2026-08-10  | 103 | 20            |
| 105 | omar       | med             | 2026-09-22  |     |               |

**Question 12**

Suppose I now want to join my `customers` dataset with the new `prices` dataset. Write out what the output would be for each type of join below.

**Question 12a**

```
full_join(customers, prices)
```

**Question 12b**

```
inner_join(customers, prices)
```

**Question 12c**

```
left_join(customers, prices)
```